

Lecture 9 2026-09-29

Where did the Lagrangian come from?

Recall NCO for equality constrained opt:

$$\nabla f(x^*) + y_i \nabla g_i(x^*) = 0 \rightarrow \mathcal{L} = f(x) + y_i g_i(x)$$

Alternatively: we can consider the Lagrangian as penalizing constraint violation

$$\text{IPM: wanted } s_i \geq 0 \rightarrow -\log(s_i)$$

Suppose our original problem is:

$$f(x)$$

$$\min_x f(x)$$

$$\text{subj. to: } g_i(x) \leq 0$$

$\Rightarrow$ we could penalize violations of $g_i(x)$

$$\text{indicator function: } \mathcal{I}_+(g_i(x)) = \begin{cases} 0 & \text{if } g_i(x) \leq 0 \\ \infty & \text{otherwise} \end{cases}$$



In penalty form, we minimize: $\min_x f(x) + \mathcal{I}_+(g_i(x))$

But this function is discontinuous and not possible to differentiate

$\rightarrow$ penalize constraint violations linearly:

$$\mathcal{L}(x, y_i) = f(x) + y_i g_i(x)$$

$$y_i \geq 0$$

With multiple constraints:

$$\mathcal{L}(x, y) = f(x) + \sum_{i=1}^P y_i g_i(x)$$

$$y \geq 0$$

Note: $\mathcal{I}_+(g(x))$ can be recovered by maximizing y :

$$\mathcal{I}_+(g(x)) = \max_{\lambda \geq 0} \mathcal{L}(x, y)$$

$$I_+(g(x)) = \max_{\lambda \geq 0} \mathcal{L}(x, \lambda)$$

Primal problem: $\min_x \mathcal{L}(x, y) \rightarrow \min_x \max_{y \geq 0} \mathcal{L}(x, y)$

$\Rightarrow$ suppose the primal attains value $\mathcal{P}^*$

Note that choice of y_i trades off minimizing $f(x)$ with penalizing constraint violation of $g_i(x)$

What if we exchanged order?

Dual problem: $\max_{y \geq 0} \min_x \mathcal{L}(x, y) \Rightarrow$ suppose this attains value $\mathcal{D}^*$

How can we relate $\mathcal{P}^*$ and $\mathcal{D}^*$?

Theorem: min-max inequality states

$$\max_x \min_y f(x, y) \leq \min_x \max_y f(x, y)$$

Recall: $\mathcal{P}^* = \min_x \max_{y \geq 0} \mathcal{L}(x, y)$

$$\mathcal{D}^* = \max_{y \geq 0} \min_x \mathcal{L}(x, y)$$

Applying min-max theorem $\rightarrow D^* \leq P^*$

$\rightarrow$ This difference between P^* and D^* is the duality gap

Strong duality! under certain conditions (e.g., Slater's conditions), we might be able to show $P^* = D^*$

Why is duality something useful to reason about?

1. The dual is always convex (we won't prove this for now) $\rightarrow$ solving for D^* may be much easier than solving for P^* and can provide an informative lower bound
2. If strong duality holds, we can just solve for D^* directly